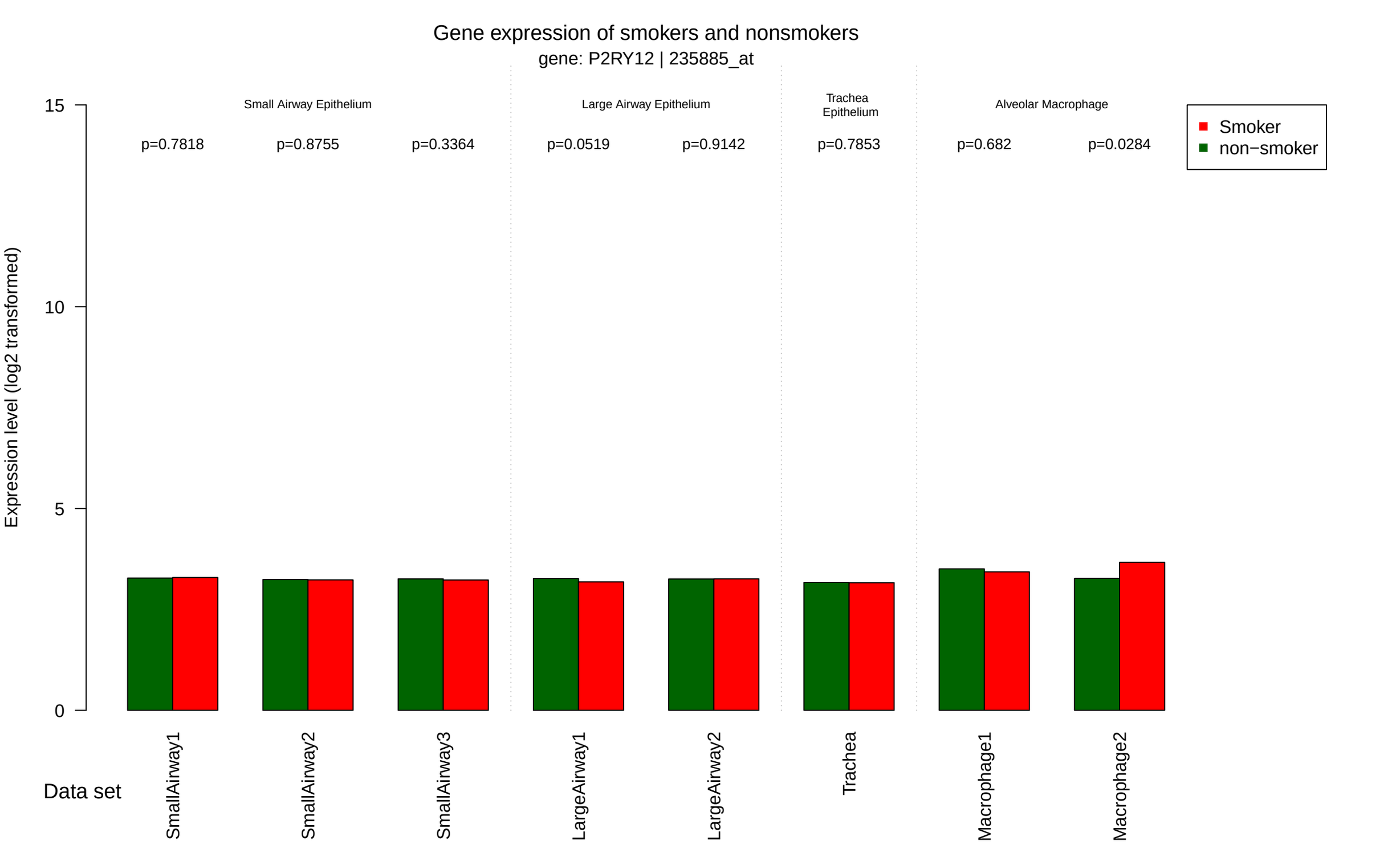
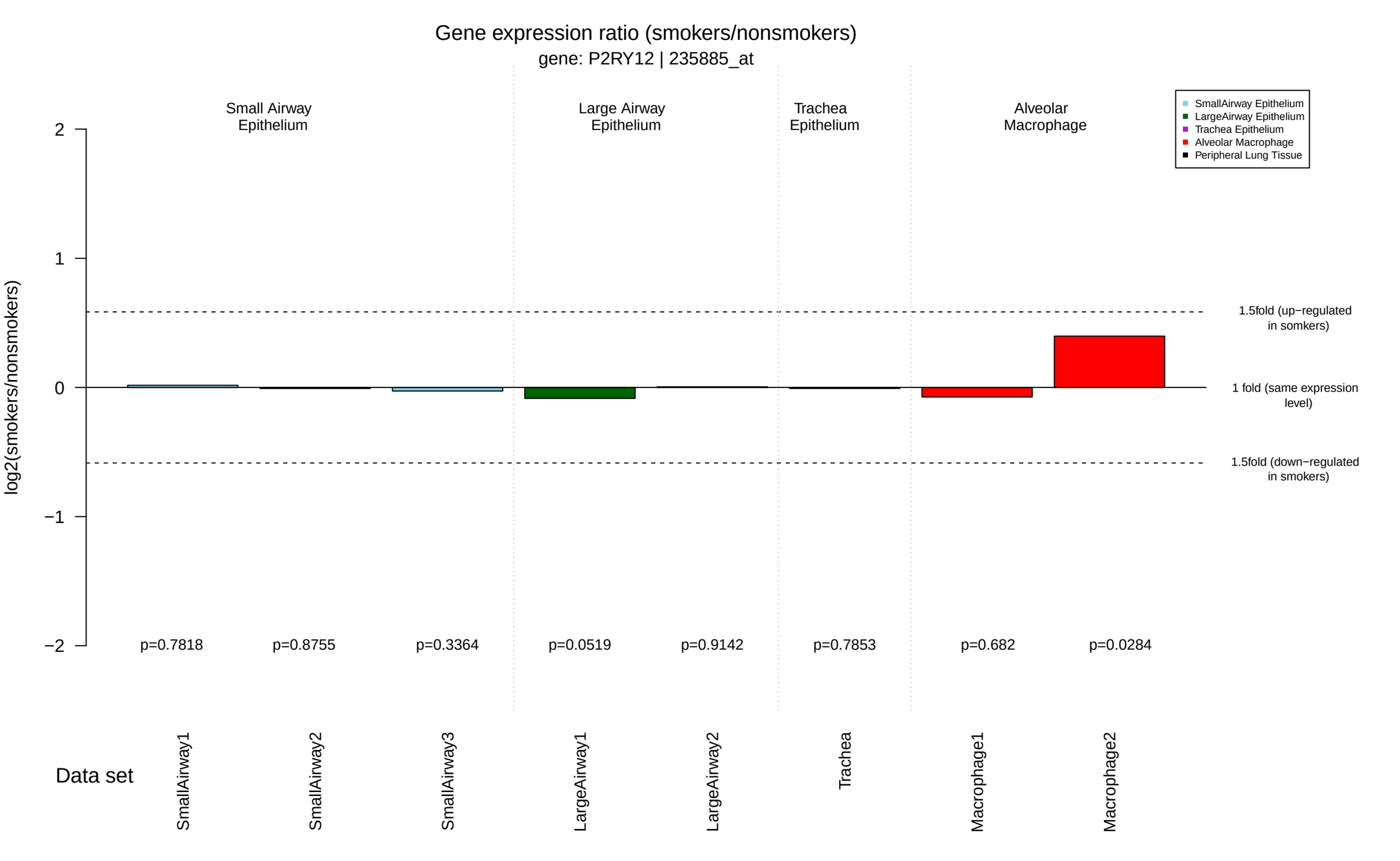
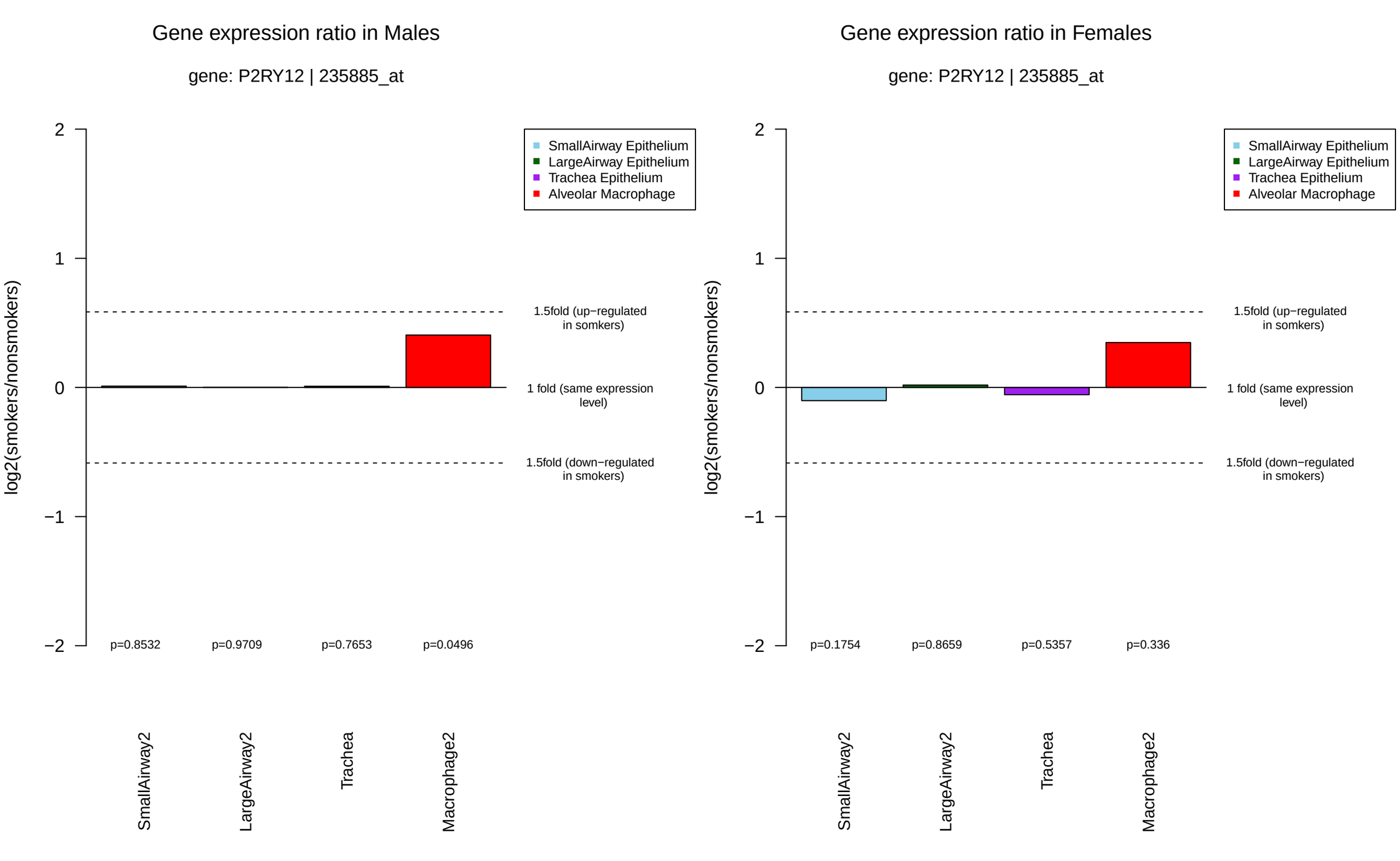
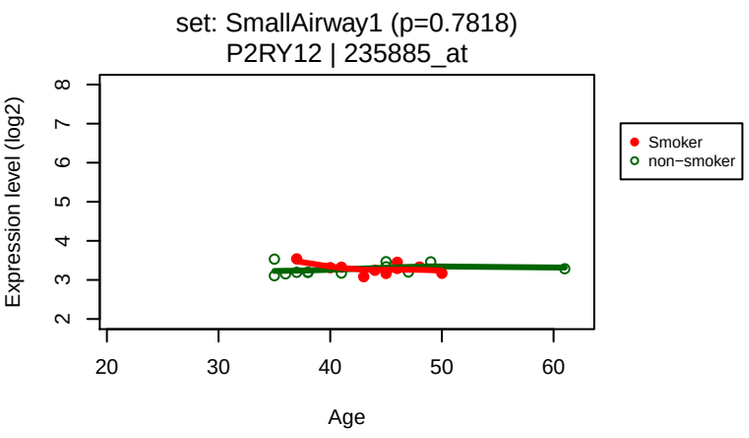
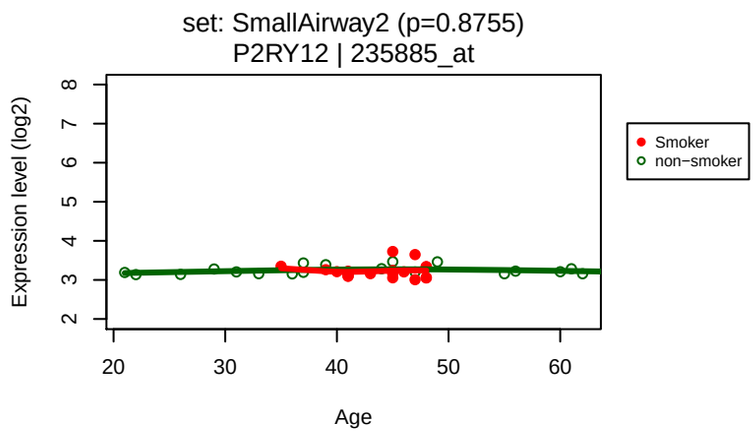
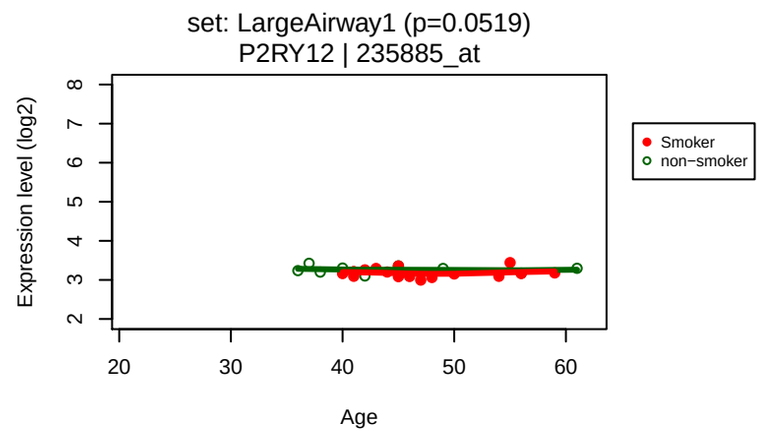

Gene expression of smokers and nonsmokers
gene: P2RY12 | 235885_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7818
p=0.8755
p=0.3364
p=0.0519
p=0.9142
p=0.7853
p=0.682
p=0.0284
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

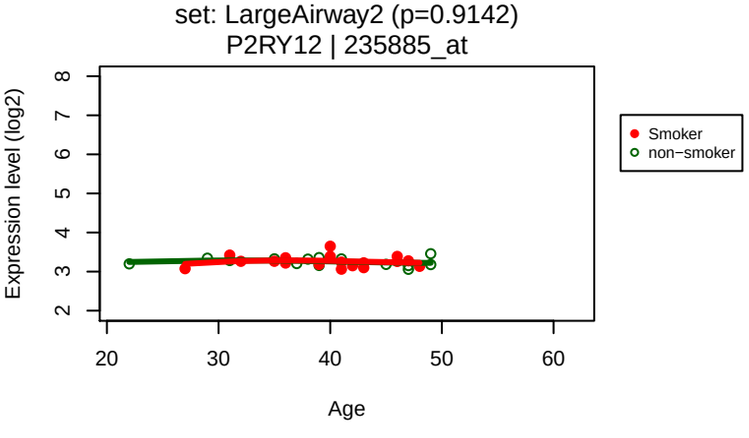

Gene expression ratio (smokers/nonsmokers)
gene: P2RY12 | 235885_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7818
p=0.8755
p=0.3364
p=0.0519
p=0.9142
p=0.7853
p=0.682
p=0.0284
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

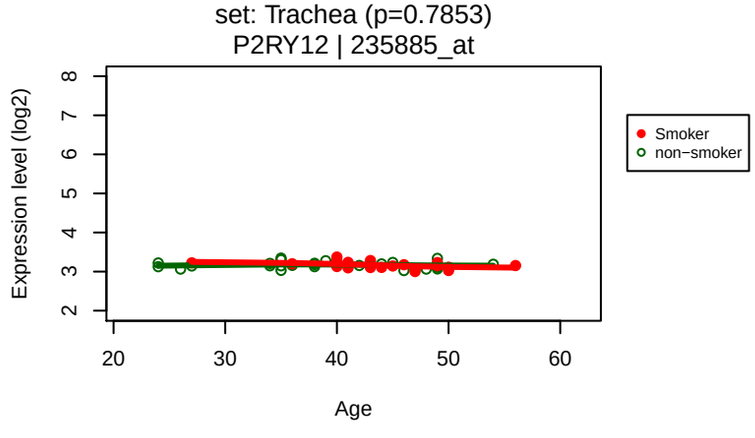

Gene expression ratio in Males
gene: P2RY12 | 235885_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8532
p=0.9709
p=0.7653
p=0.0496
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: P2RY12 | 235885_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1754
p=0.8659
p=0.5357
p=0.336
SmallAirway2
LargeAirway2
Trachea
Macrophage2

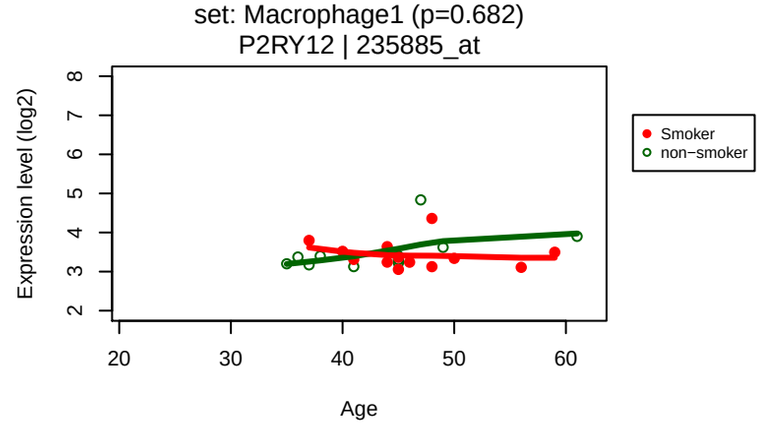

set: SmallAirway1 (p=0.7818)
P2RY12 | 235885_at
Expression level (log2)
Age
Smoker
non−smoker

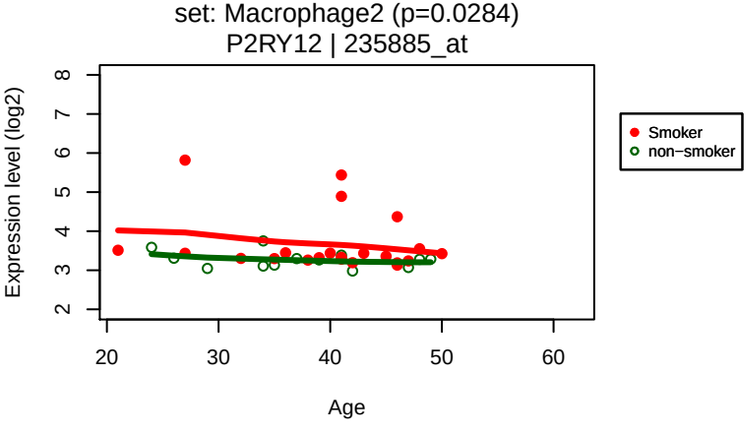

set: SmallAirway2 (p=0.8755)
P2RY12 | 235885_at
Expression level (log2)
Age
Smoker
non−smoker

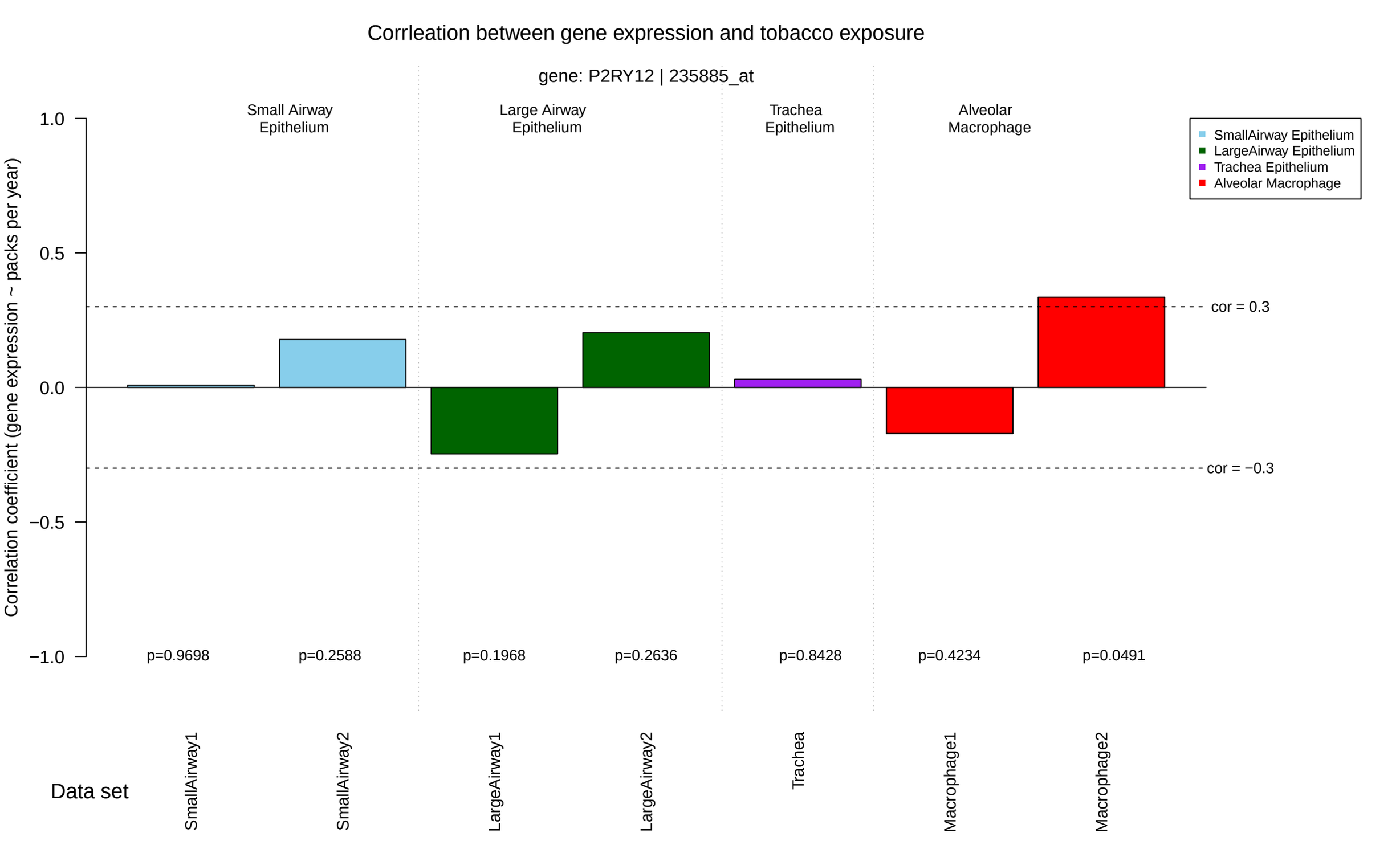

set: LargeAirway1 (p=0.0519)
P2RY12 | 235885_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9142)
P2RY12 | 235885_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7853)
P2RY12 | 235885_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.682)
P2RY12 | 235885_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0284)
P2RY12 | 235885_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: P2RY12 | 235885_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9698
p=0.2588
p=0.1968
p=0.2636
p=0.8428
p=0.4234
p=0.0491
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2